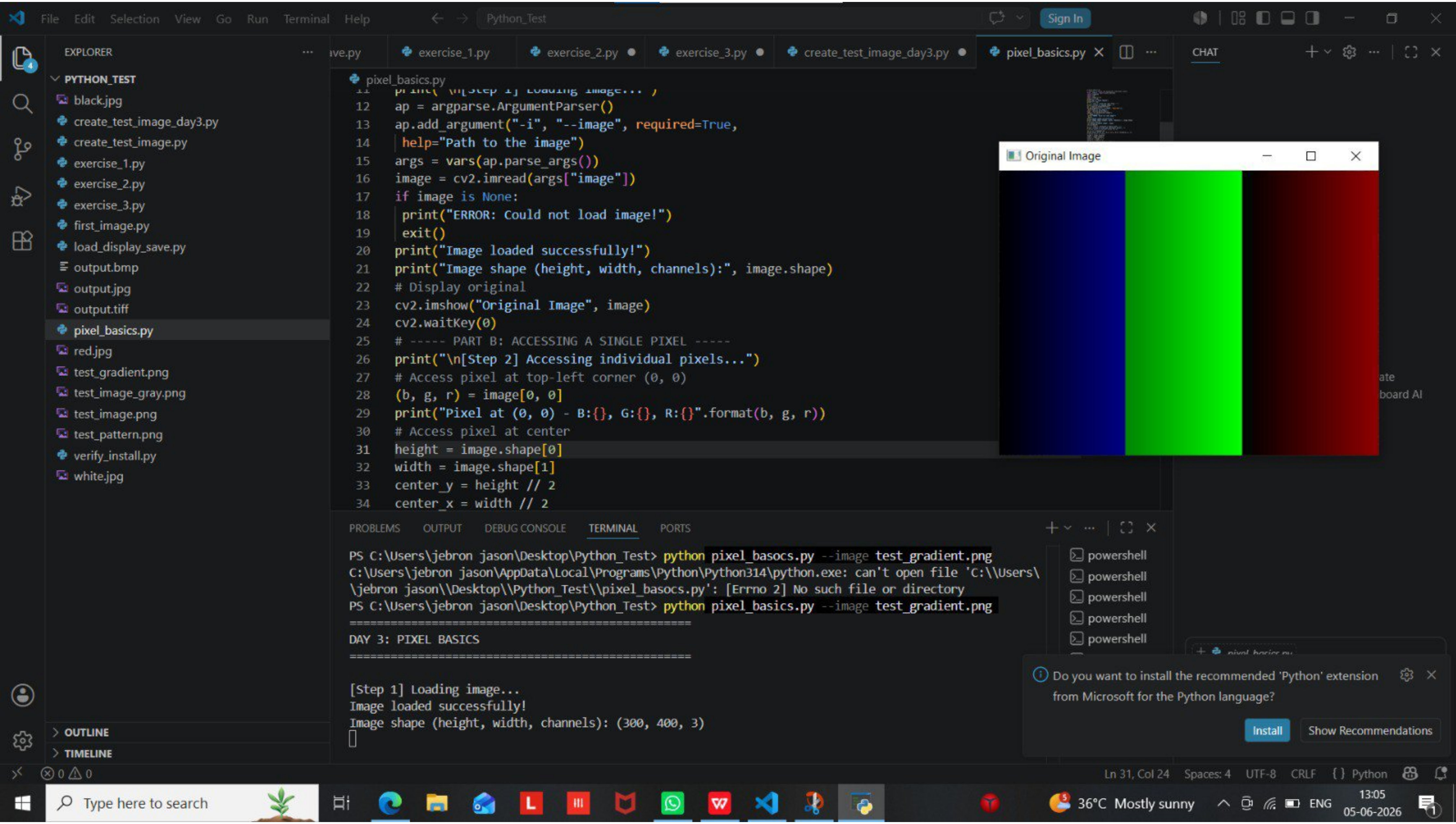
```python
# create_test_image_day3.py

import numpy as np
import cv2

# Create a gradient image to see pixel values clearly
image = np.zeros((300, 400, 3), dtype="uint8")

# Fill with gradient
for y in range(300):
    for x in range(400):

        # Different colors for different regions
        if x < 133:
            image[y, x] = (x % 256, 0, 0)  # Blue gradient

        elif x < 266:
            image[y, x] = (0, x % 256, 0)  # Green gradient

        else:
            image[y, x] = (0, 0, x % 256)  # Red gradient

cv2.imwrite("test_gradient.png", image)
```

```
PS C:\Users\jebron jason\Desktop\Python_Test> python create_test_image_day3.py
Created test_gradient.png
Created test_pattern.png
PS C:\Users\jebron jason\Desktop\Python_Test> python create_test_image_day3.py
Created test_gradient.png
Created test_pattern.png
PS C:\Users\jebron jason\Desktop\Python_Test>
```


File Edit Selection View Go Run Terminal Help
Python_Test
Sign In
EXPLORER
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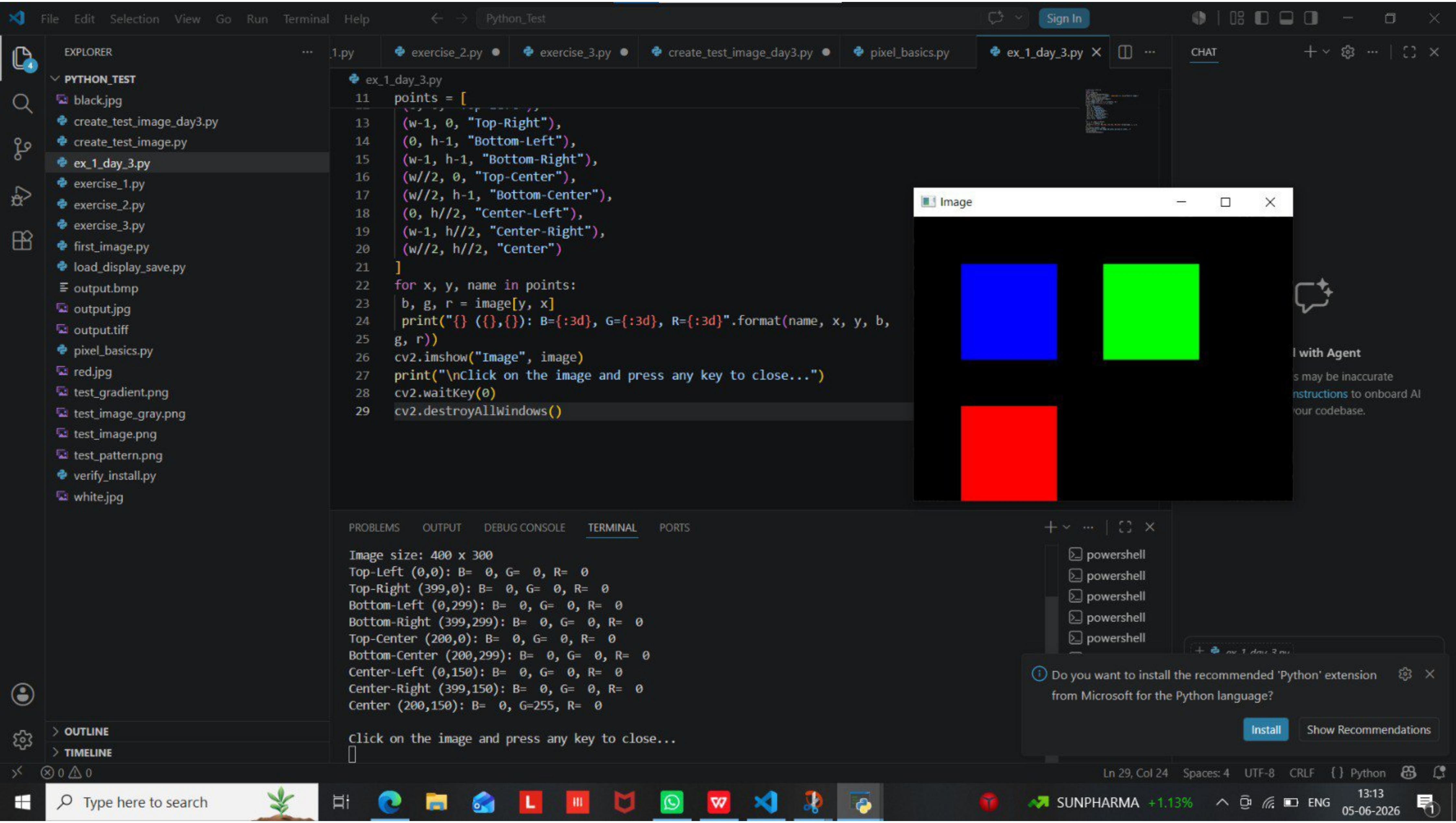
CHAT
ap = argparse.ArgumentParser()
ap.add_argument("-i", "--image", required=True,
help="Path to the image")
args = vars(ap.parse_args())
image = cv2.imread(args["image"])
if image is None:
print("ERROR: Could not load image!")
exit()
print("Image loaded successfully!")
print("Image shape (height, width, channels):", image.shape)
# Display original
cv2.imshow("Original Image", image)
cv2.waitKey(0)
# ----- PART B: ACCESSING A SINGLE PIXEL -----
print("\n[Step 2] Accessing individual pixels...")
# Access pixel at top-left corner (0, 0)
(b, g, r) = image[0, 0]
print("Pixel at (0, 0) - B:{}, G:{}, R:{}".format(b, g, r))
# Access pixel at center
height = image.shape[0]
width = image.shape[1]
center_y = height // 2
center_x = width // 2
Original Image
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\jebron jason\Desktop\Python_Test> python pixel_basocs.py --image test_gradient.png
C:\Users\jebron jason\AppData\Local\Programs\Python\Python314\python.exe: can't open file 'C:\\Users\\jebron jason\\Desktop\\Python_Test\\pixel_basocs.py': [Errno 2] No such file or directory
PS C:\Users\jebron jason\Desktop\Python_Test> python pixel_basics.py --image test_gradient.png
==
DAY 3: PIXEL BASICS
==
[Step 1] Loading image...
Image loaded successfully!
Image shape (height, width, channels): (300, 400, 3)
powershell
powershell
powershell
powershell
powershell
Do you want to install the recommended 'Python' extension from Microsoft for the Python language?
Install
Show Recommendations
Ln 31, Col 24 Spaces: 4 UTF-8 CRLF {} Python
Type here to search
36°C Mostly sunny
ENG
13:05
05-06-2026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TIMELINE
1.py
exercise_2.py
exercise_3.py
create_test_image_day3.py
pixel_basics.py
ex_1_day_3.py
points = [
(w-1, 0, "Top-Right"),
(0, h-1, "Bottom-Left"),
(w-1, h-1, "Bottom-Right"),
(w//2, 0, "Top-Center"),
(w//2, h-1, "Bottom-Center"),
(0, h//2, "Center-Left"),
(w-1, h//2, "Center-Right"),
(w//2, h//2, "Center")
]
for x, y, name in points:
b, g, r = image[y, x]
print("{} ({},{}): B={:3d}, G={:3d}, R={:3d}".format(name, x, y, b,
g, r))
cv2.imshow("Image", image)
print("\nClick on the image and press any key to close...")
cv2.waitKey(0)
cv2.destroyAllWindows()
Image
CHAT
with Agent
s may be inaccurate
nstructions to onboard AI
our codebase.
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Image size: 400 x 300
Top-Left (0,0): B= 0, G= 0, R= 0
Top-Right (399,0): B= 0, G= 0, R= 0
Bottom-Left (0,299): B= 0, G= 0, R= 0
Bottom-Right (399,299): B= 0, G= 0, R= 0
Top-Center (200,0): B= 0, G= 0, R= 0
Bottom-Center (200,299): B= 0, G= 0, R= 0
Center-Left (0,150): B= 0, G= 0, R= 0
Center-Right (399,150): B= 0, G= 0, R= 0
Center (200,150): B= 0, G=255, R= 0
Click on the image and press any key to close...
powershell
powershell
powershell
powershell
powershell
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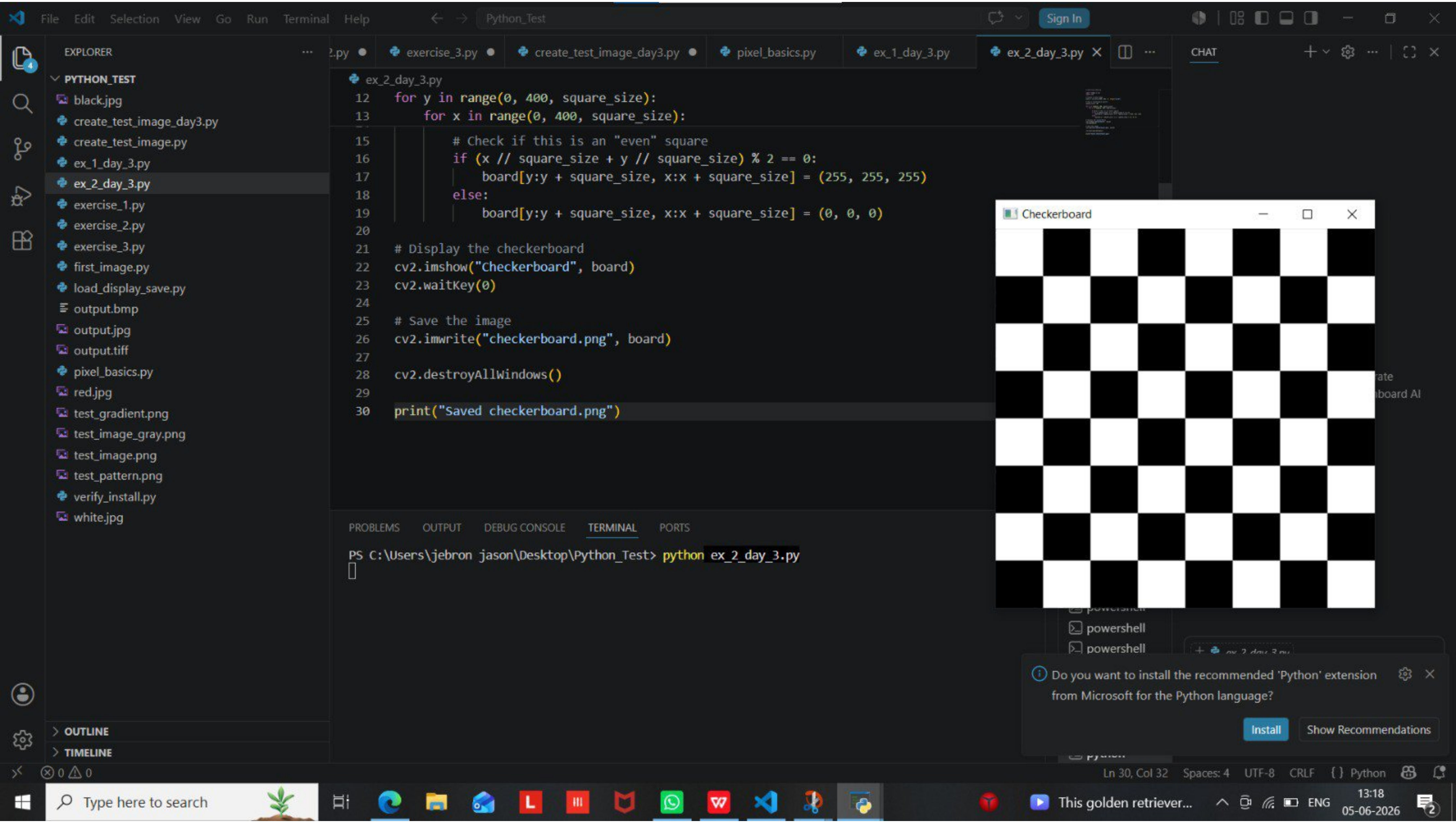
Ln 29, Col 24 Spaces: 4 UTF-8 CRLF {} Python
Type here to search
SUNPHARMA +1.13%
ENG
13:13
05-06-2026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test_image.png
test_pattern.png
verify_install.py
white.jpg
OUTLINE
TIMELINE
exercise_3.py
create_test_image_day3.py
pixel_basics.py
ex_1_day_3.py
ex_2_day_3.py
CHAT
for y in range(0, 400, square_size):
for x in range(0, 400, square_size):
# Check if this is an "even" square
if (x // square_size + y // square_size) % 2 == 0:
board[y:y + square_size, x:x + square_size] = (255, 255, 255)
else:
board[y:y + square_size, x:x + square_size] = (0, 0, 0)
# Display the checkerboard
cv2.imshow("Checkerboard", board)
cv2.waitKey(0)
# Save the image
cv2.imwrite("checkerboard.png", board)
cv2.destroyAllWindows()
print("Saved checkerboard.png")
Checkerboard
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\jebron jason\Desktop\Python_Test> python ex_2_day_3.py
powershell
powershell
Do you want to install the recommended 'Python' extension from Microsoft for the Python language?
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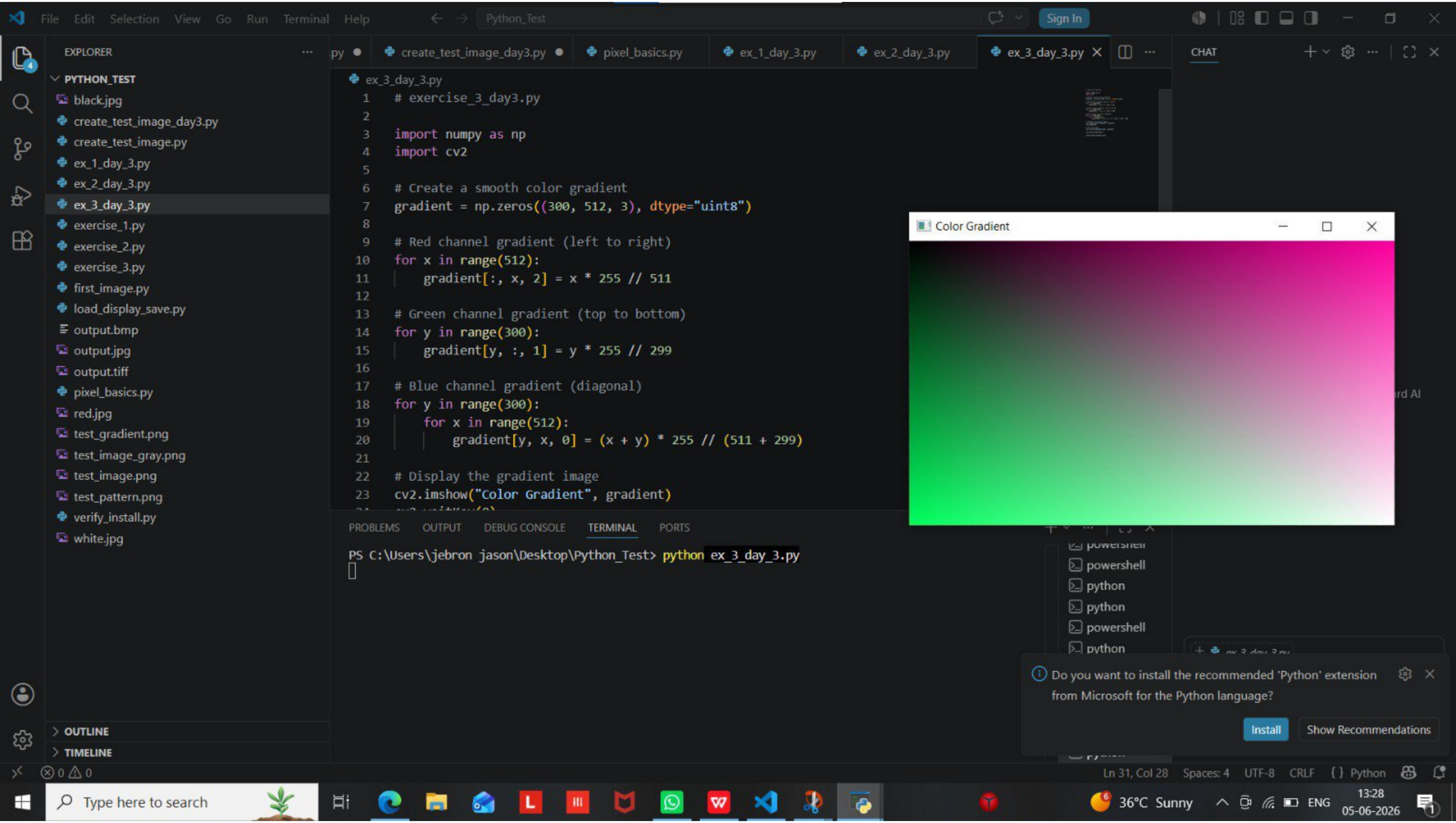
Ln 30, Col 32 Spaces: 4 UTF-8 CRLF { } Python
Type here to search
This golden retriever...
ENG
13:18
05-06-2026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CHAT
# exercise_3_day3.py
import numpy as np
import cv2
# Create a smooth color gradient
gradient = np.zeros((300, 512, 3), dtype="uint8")
# Red channel gradient (left to right)
for x in range(512):
gradient[:, x, 2] = x * 255 // 511
# Green channel gradient (top to bottom)
for y in range(300):
gradient[y, :, 1] = y * 255 // 299
# Blue channel gradient (diagonal)
for y in range(300):
for x in range(512):
gradient[y, x, 0] = (x + y) * 255 // (511 + 299)
# Display the gradient image
cv2.imshow("Color Gradient", gradient)
Color Gradient
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\jebron jason\Desktop\Python_Test> python ex_3_day_3.py
powershell
python
python
powershell
python
Do you want to install the recommended 'Python' extension from Microsoft for the Python language?
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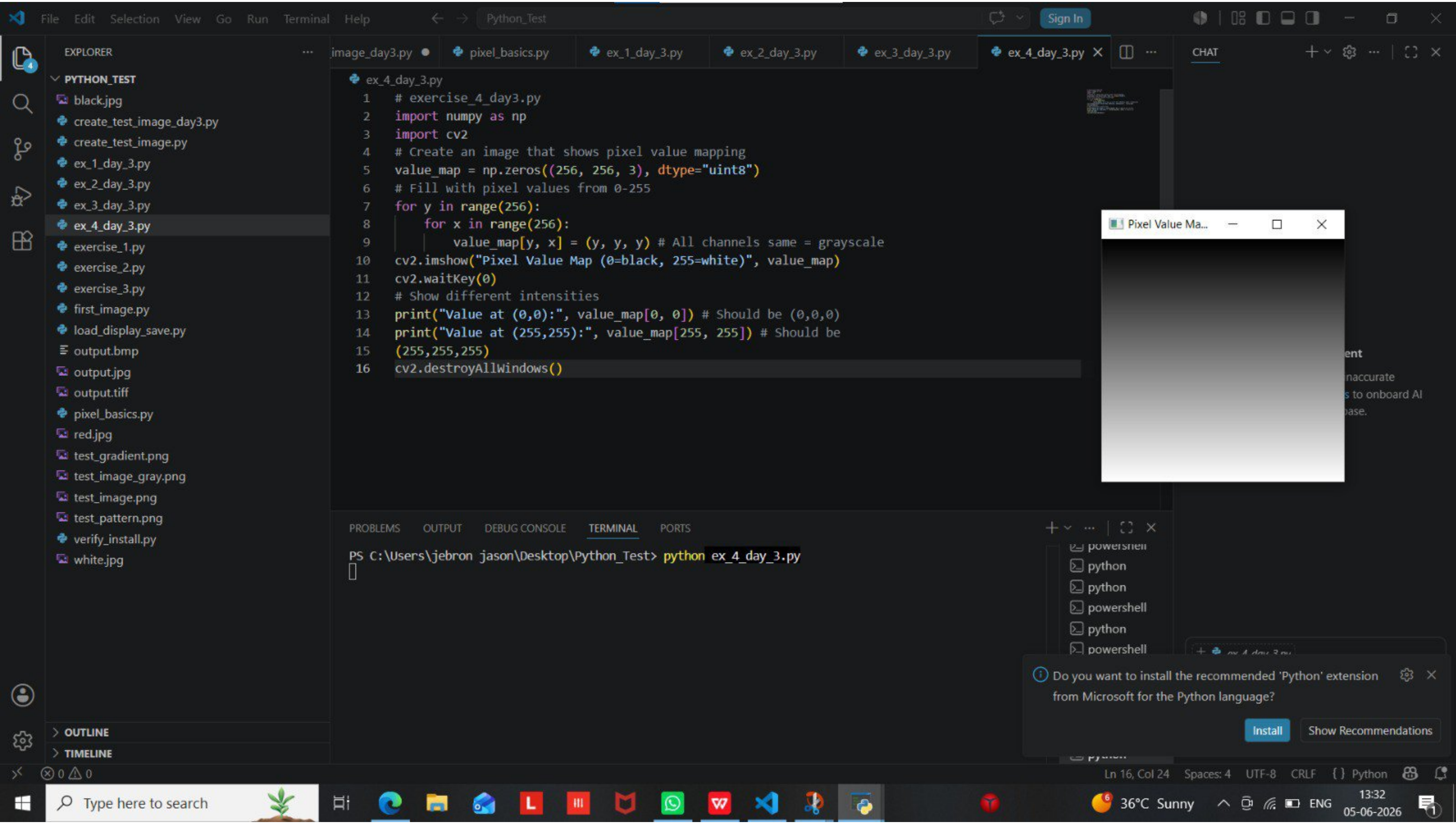
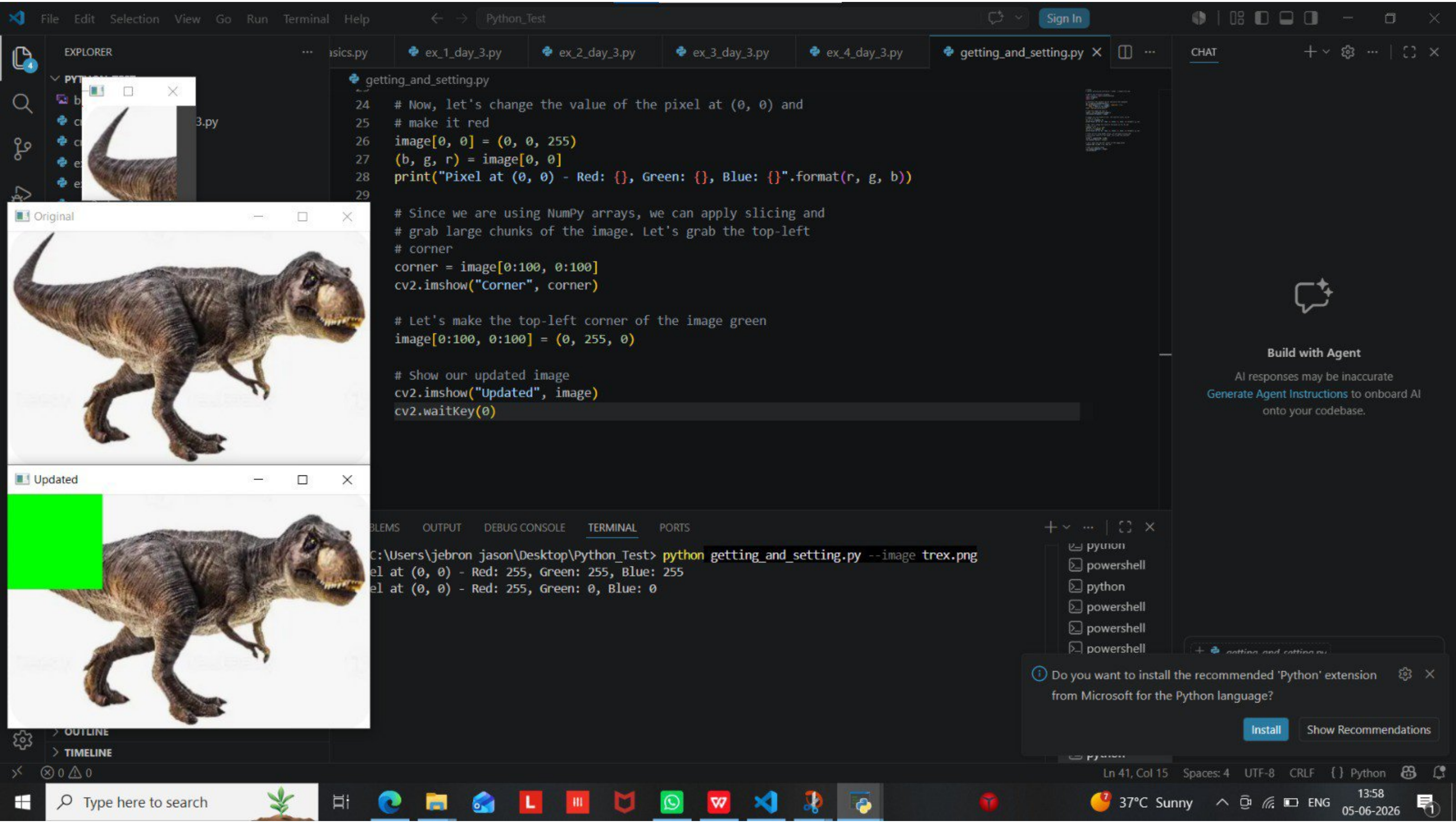
Show Recommendations
Ln 31, Col 28 Spaces: 4 UTF-8 CRLF { } Python
Type here to search
36°C Sunny
ENG
13:28
05-06-2026



ex_4_day_3.py
# exercise_4_day3.py
import numpy as np
import cv2
# Create an image that shows pixel value mapping
value_map = np.zeros((256, 256, 3), dtype="uint8")
# Fill with pixel values from 0-255
for y in range(256):
    for x in range(256):
        value_map[y, x] = (y, y, y) # All channels same = grayscale
cv2.imshow("Pixel Value Map (0=black, 255=white)", value_map)
cv2.waitKey(0)
# Show different intensities
print("Value at (0,0):", value_map[0, 0]) # Should be (0,0,0)
print("Value at (255,255):", value_map[255, 255]) # Should be
(255,255,255)
cv2.destroyAllWindows()
PS C:\Users\jebron jason\Desktop\Python_Test> python ex_4_day_3.py
Pixel Value Ma...
Do you want to install the recommended 'Python' extension from Microsoft for the Python language?
Install
Show Recommendations



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EXPLORER
ex_1_day_3.py
ex_2_day_3.py
ex_3_day_3.py
ex_4_day_3.py
getting_and_setting.py
CHAT
Now, let's change the value of the pixel at (0, 0) and
make it red
image[0, 0] = (0, 0, 255)
(b, g, r) = image[0, 0]
print("Pixel at (0, 0) - Red: {}, Green: {}, Blue: {}".format(r, g, b))
# Since we are using NumPy arrays, we can apply slicing and
# grab large chunks of the image. Let's grab the top-left
# corner
corner = image[0:100, 0:100]
cv2.imshow("Corner", corner)
# Let's make the top-left corner of the image green
image[0:100, 0:100] = (0, 255, 0)
# Show our updated image
cv2.imshow("Updated", image)
cv2.waitKey(0)
Original
Updated
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
C:\Users\jebron jason\Desktop\Python_Test> python getting_and_setting.py --image trex.png
el at (0, 0) - Red: 255, Green: 255, Blue: 255
el at (0, 0) - Red: 255, Green: 0, Blue: 0
powershell
python
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
Do you want to install the recommended 'Python' extension from Microsoft for the Python language?
Install
Show Recommendations
OUTLINE
TIMELINE
Ln 41, Col 15
Spaces: 4
UTF-8
CRLF
Python
Type here to search
37°C Sunny
ENG
13:58
05-06-2026